

# (English/Arabic) Grammar for Requirement Engineers

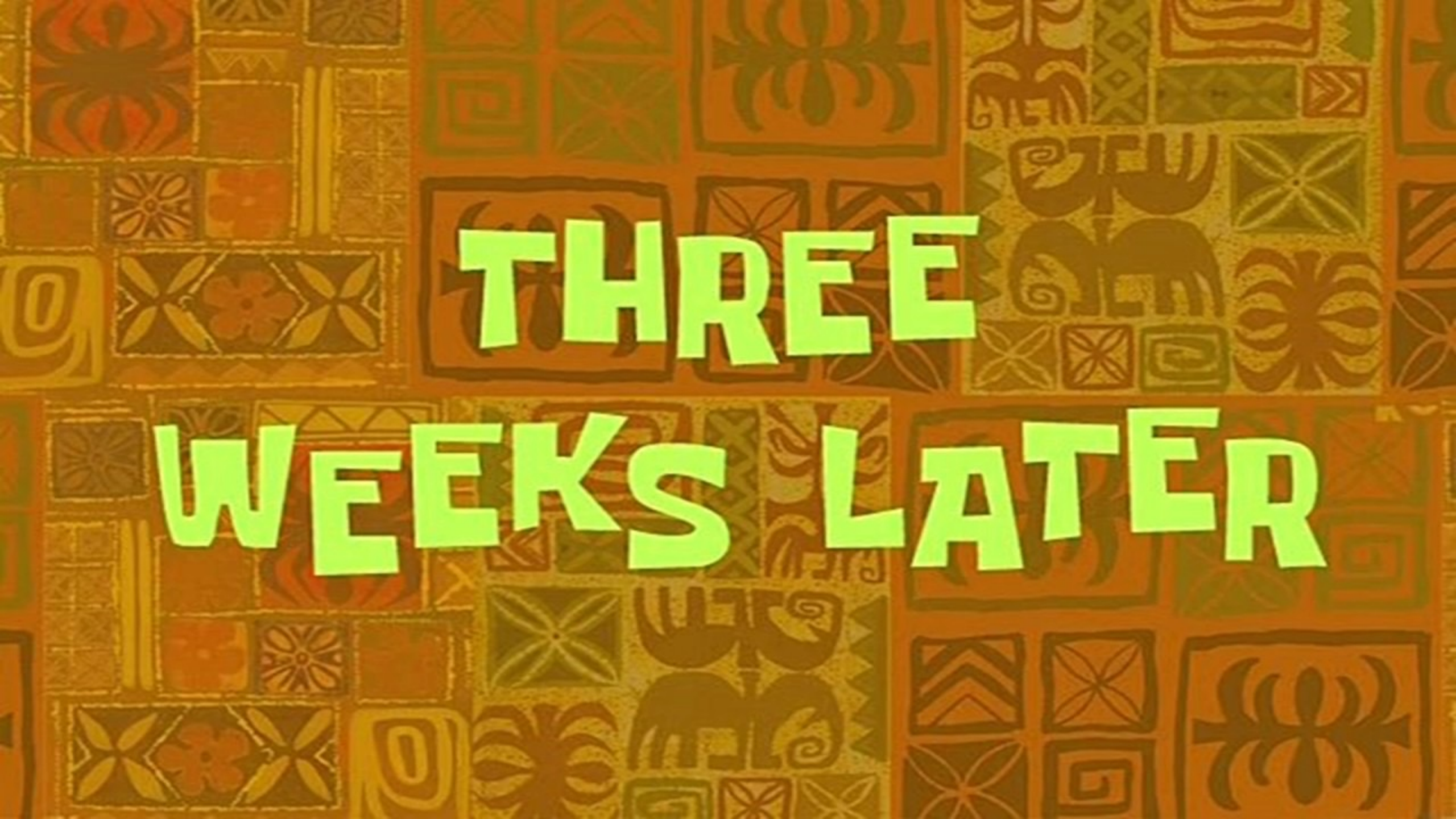
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# Background

We want every user  
to have an **account**  
in the bank system.

نريد أن يكون لكل  
مستخدم حساب في النظام



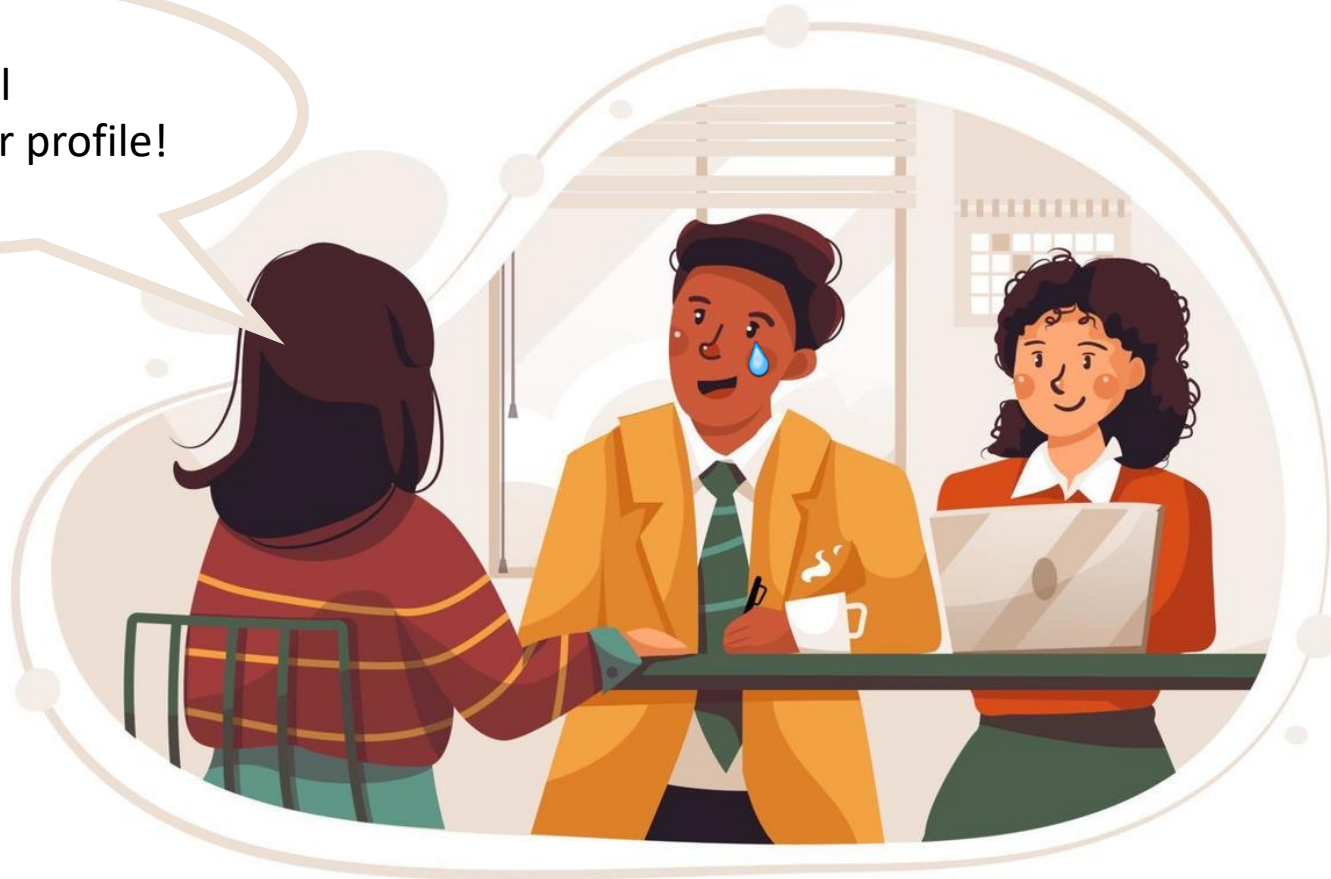


**THREE  
WEEKS LATER**



# Background

I meant a financial  
account not a user profile!



# Problem

- In requirements elicitation, clarifying domain-specific meaning is essential.
- Misunderstandings lead to iterations, becomes costly if detected late.
- Problem with existing tools:
  - Google translate → Wrong grammar leads to wrong interpretation.
  - LLMs → general vocabulary leads to confusion.

# Solution

Using GF to:

- Create a domain-specific lexicon.
- Grammar tailored to requirement writing style
  - e.g., “As a <stakeholder>, I want to <feature>.
- English-Arabic requirements translation.

```
-- Roles
Student  = mkRole "طالب" ;
Customer = mkRole "زبون" ;
User     = mkRole "مستخدم" ;
Manager  = mkRole "مدير" ;
Developer = mkRole "مبرمج" ;
Tester   = mkRole "مختبر" ;
ProjectManager = mkRole (nisbaA "مشروع") (mkN "مدير") ;
EndUser  = mkRole (nisbaA "نهائي") (mkN "مستخدم") ;
Administrator = mkRole "مسؤول" ;
OnlineCustomer = mkRole (nisbaA "إلكتروني") (mkN "زبون") ;
```

```
-- actions
Read = mkVP <LexiconAra.read_V2 : V> ;
Search = mkVP (mkV "يُبحَثُ" "بَحْث") ;
Merge = mkV2 (mkV "يُدمَجُ" "دَمَج") ;
Download = mkV2 (mkV "يُحمَلُ" "حَمَل") ;
View = mkV2 (mkV "يُزَي" "رُؤْيَة") ;
Edit = dirV2 (v2 "عدل") ;
Generate = dirV2 (v4 "نشأ") ;
Log = dirV2 (v2 "سجل") ;
Assign = dirV2 (v2 "عين") ;
Create = dirV2 (v4 "نشأ") ;
Track = mkV2 (v3 "تبع") ;
Update = dirV2 (v2 "حدث") ;
```

```
-- things
File = mkThing "ملف" ;
Code = mkThing "شيفرة" ;
Account = mkThing "حساب" ;
Order = mkThing "طلب" ;
CourseMaterials = mkThing (nisbaA "دُرس") (mkN "مواد") ;
AccountInformation = mkThing (nisbaA "حساب") (mkN "معلومات") ;
Report = mkThing "تقرير" ;
PurchaseHistory = mkThing (nisbaA "مشتريات") (mkN "تاريخ") ;
```

# Solution

Using GF to:

- Create a domain-specific lexicon.
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oper

```
-- As a Person, ...  
as_a_NP_Adv : NP -> Adv =  
  \np -> mkAdv kaPrep np ;
```

lin

```
SAsRole role actionVP =  
  mkS (as_a_NP_Adv role)  
  | (mkS (mkCl i_NP (mkVP want_VV actionVP))) ;  
  
SAsRoleV2 role doSomething thing =  
  mkS (as_a_NP_Adv role)  
  | (mkS (mkCl i_NP want_VV (Apply doSomething thing))) ;
```

# Solution

```
App> p "as an administrator I want to create an account"  
كَمَسْؤُولٍ أُرِيدُ أَنْ أُنْشِئَ حِسَاباً
```

Using GF to:

```
App> p "as a customer I want to track an order"  
كَزْبُونٍ أُرِيدُ أَنْ أَتَابِعَ طَلَباً
```

- Create a domain-specific lexicon.
- Grammar tailored to requirement writing style
  - e.g., “As a <stakeholder>, I want to <feature>.”
- English-Arabic requirements translation.



# Next steps

- Punctuations
- Extend the grammar
  - “As a <stakeholder>, I want to <feature>, **in order to <reason>**”
- Wrap the program in Python

